

CATTLE FEEDER INITIATIVE

US Department of Agriculture | Office of the Chief Information Officer

Daniel Rhee, Software Engineering Fellow | University of California, Santa Cruz, *Computer Science BS*

Keywords:

Computer Vision, Explainable Artificial Intelligence, Data Augmentation

Summary:

To increase both state and national level understanding of fair market cattle prices, Daniel worked with the US Department of Agriculture's Office of the Chief Information Officer to validate the possibility of doing cattle sex classification using computer vision. Utilizing PyTorch and Databricks, Daniel was able to achieve high accuracy on a human interpretable prediction model. This validation could potentially lead to the deployment of cattle sex prediction models on edge system at more cattle auctions and provide the USDA with more data to assess fair market price.

Cattle Feeder Initiative

Office of the Chief Information Officer
Cattle Feeder Project

Robert Eaton — Director - Livestock, Poultry, and Grain Market News
Division

coding it forward >



Daniel Rhee

CATTLE SALES

- Cattle is bought and sold many times before it is processed and products are consumer ready
- Cattle with different weight, sex, and muscle mass are all priced differently
- The government currently has gaps in coverage to best understand pricing



33,600,000

Feeder Cattle are available to be bought and sold in the United States

CURRENT USDA INVOLVEMENT

- The Agricultural Marketing Service (AMS) by the USDA aims to provide market data on food (including cattle)
- Trained and tested AMS reporters log frame size, muscle thickness, and sex/class of cattle sold on the ground at auctions
- AMS takes data across many auctions and compiles daily or weekly reports on regional and national pricing for cattle to provide fair market value estimates

FRAME		
Expected Weight to Grade Choice		
<u>FRAME</u>	<u>STEERS</u>	<u>HEIFERS</u>
Large +		
L Large	↑	↑
Large -	1250#	1150#
Medium +		
M Medium	↑	↑
Medium -	1100#	1000#
Small +	1100#	1000#
S Small	↓	↓
Small -		

MUSCLE SCORE	
<u>MINIMUM THICKNESS</u>	<u>DEGREE OF THICKNESS</u>
1	Moderately thick -
2	Tends to be Slightly Thick -
3	Thin -
4	

WHY

- Grades provide simple and understandable ‘buckets’ to grade cattle into, allowing every farmer to understand them
- Public data allows producers and buyers to benchmark their sale and purchase prices against regional and national sales
- Trends can be observed with pricing, allowing public policy and industry to anticipate changes and adjust

CURRENT PROBLEMS

- Not all auctions in all states can be covered
- AMS reporters require training and also occasionally differ in evaluation
- Reporter shortages often leave reporters spread thin across different auctions, creating poor coverage



A Computer Vision Solution

COMPUTER VISION

- Computer Vision can allow a computer system to learn from labeled examples and perform classification without a human
- Interpretable Artificial Intelligence, like GradCam, can still allow trust to be built with human understanding to the computer's decisions
- A lower cost and permanent computer system can be installed with a camera that monitors cattle and can classify them as they move to be sold and report data back to the USDA

TRAINING

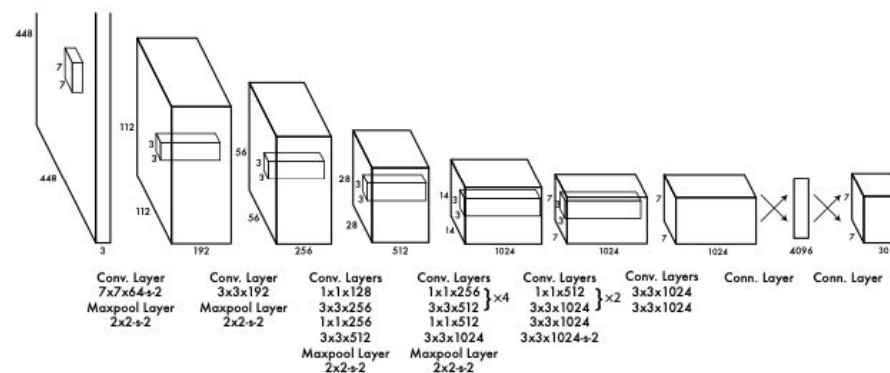
- Raw dataset had 20k+ raw images from cattle at different feed lots
- The filtered dataset had closer to 2k images after many were filtered out as low quality datapoints or duplicates
- The data was then augmented with rotations, mirrors, crops, and then greyscaled
- A CNN classifier taking a 512x512 input was then trained to classify the data

INFERENCE STAGES

1. Identify if cattle is present in the image
2. Create a segmentation box around the cattle if it is present and crop into just the animal
3. Classify the sex of the animal
4. Report confidence in the guess and create an interpretable attention map to understand the model's prediction

INFERENCE STAGES 1 & 2

1. An image is fed into the model pipeline
2. The image is run through the YOLOv8 model trained on MS COCO to identify if cattle is present, and if one is, it will be digitally segmented
3. The model will determine if the image is good enough to attempt an inference run on by measuring its blurriness using a blur kernel



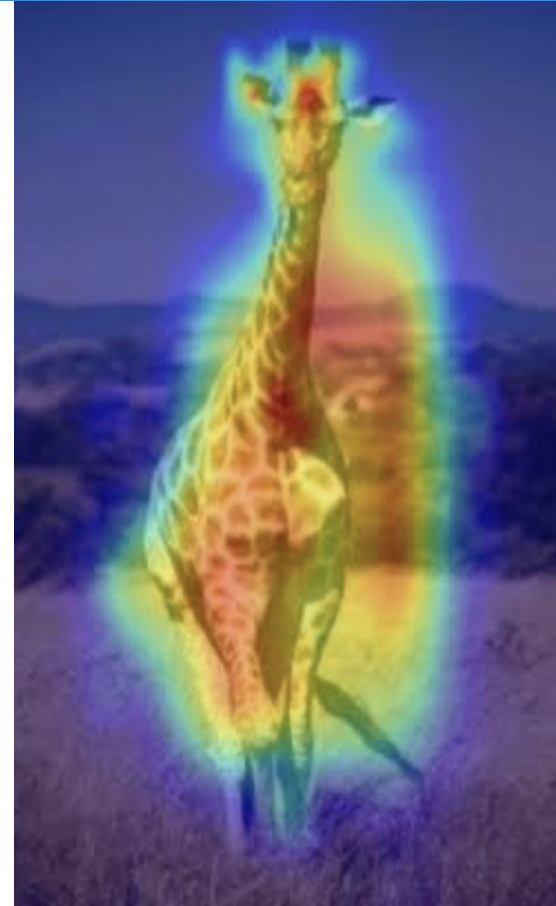
INFERENCE STAGE 3

1. The cropped image is padded, scaled to the correct input dimensions, 512 x 512
2. The CNN Classifier predicts the sex of the cattle using four 2D convolution layers
3. The output of the classifier is read along with its confidence in the prediction



INFERENCE STAGE 4

- The confidence of the prediction is drawn directly from the output of the neural network
- An attention map is created with GradCam which examines the gradients of the final convolution layer to understand where the model is focusing on the most for its identification
- A human interpretable graphic is created and outputted



RESULTS

- A data classifier with over 89% accuracy in predicting the sex of an animal when compared with human generated labels, however not always focused on the cattle
- The model often learned from the surrounding environment, such as from the carpet and pipes in the background
- A highly confident model that does not produce purely arbitrarily confidence values
- Validation that it is likely possible for computer vision to accurately classify the sex of cattle even with a simple lightweight model

NEXT STEPS

- Increasing model focus on the cattle and not the surrounding environment
- Developing a more interpretable model
- Explicit tuning of model confidence
- Modification of the model to be accelerated and deployable on lightweight edge devices in the field





cif>

Thank You!